

One solution is to allow a coolant to flow in between the plates of the heat-sink. This coolant would absorb heat from the heat-sink and allow for a more maintained environment of operation. A common coolant in such a solution would be distilled water mixed with ethylene glycol. "Proportion of these two would form the required temperature," explains Ramanan.

Interface materials are too often forgotten

Thermal interface materials applied between the conductive layer and the heat-sink to facilitate heat transfer are too often forgotten. For efficient transfer of heat from the circuit to the sink, a proper interface becomes necessary. "The heat-sink in most cases will be rigid, hence reducing the contact area, resulting in a lower level of heat transfer," says Padmanabha Shakthivelu, general manager, Electrolube India. As a result, thermal interfaces become integral to a system.

Depending on the material, this layer can cause significant difference in the effectiveness of the heat-sink. Thermal solutions are also available in the form of adhesives, greases, gels, pads or solder alloys. Dr Misra says, "Low hardness and high conductivity are the necessary requirements for interface materials."

Some solutions are solid. Flexible solid interface materials consisting of silicon in ceramic are in high demand. However, a lot of forums raise the question of local vendors preferring one over the other, depending on

A turn of events in LEDs

A common misconception with LEDs is that these do not often release heat. LEDs are cool to touch since these usually do not produce heat in the form of infrared radiation. There is, however, a lot of unwanted heat generated within the devices. This is the heat generated due to the inefficiency of semiconductors that generate light. For reference, radiant efficiency of typical LEDs is about five per cent to 40 per cent, which translates to 95 per cent to 60 per cent of energy loss through heat.

Recently, some LEDs are being marketed with claims of no heat-sink at all. However, that is not entirely true, as the heat-sink is integrated into the device for improvement in aesthetics. You did not really think these had done away with the heat-sink altogether, did you?

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the area of usage. Thermal pads and thermal compounds are used to fill air gaps caused by imperfectly flat or smooth surfaces, which should be in thermal contact. Thermal pads are solid, rubber-like materials, often based on various silicone compounds or paraffin wax. These are relatively firm at room temperature, but become soft and are well able to fill gaps at higher temperatures.

Average usage lifetime for a solution usually goes from one to three years. However, with thermal pastes, this can be more towards the lower end of the spectrum, due to their tendency to dry out. There is always a risk of thermal grease dispersing over time, leaving no interface material between the heat source and the sink.

Others are liquid. The liquid approach offers infinite thickness variations with little to no stress to sensitive components during assembly. Gap-filler 3500LV from Henkel is a two-part, high thermal conductivity, liquid gap-filling material. Made of polymer and thermally-conductive elements, it offers mechanical property benefits of a silicone material with low outgassing.

ER2224 epoxy resin from Electrolube caters to a wide range of automotive applications. These generally require

high thermal conductivity and good thermal cycling performance.

Shakthivelu explains the design of grease solutions for heat management. "We have a range of grease, for example, one-watt, three-watt or five-watt grease." These have to be designed so that these are the perfect balance of spread and conductivity. He adds, "We have small metal conductors inside the paste, which transfer the heat. Sizes of these conductors change with the amount and rate of transfer of heat."

Where does this trend take devices

The role of thermal compound is simple—to fill the tiny gaps between the cooler and the integrated heat spreader, and to promote thermal conductivity. Some reports suggest that, introducing active cooling systems reduces the size of heat-sinks by 90 per cent. Now, if you could design a system with a smaller size, that would be a dream for designers.

Manufacturers are sceptical, however, about the reliability of fans and the noise these could create. Misra says, "Heat absorbers is an area that shows promise." Introduction of new materials and form factors has been a very welcome change in designing heat-dissipation solutions. These improvements should lead to things that have never before been possible. **EFY**



Fig. 2: Thermal pad and paste after prolonged usage